

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination – Summer 2022

Course: T.Y. B. Tech.

Branch: Mechanical Engineering

Semester: V

Subject Code & Name: BTMEC505 & Metrology and Quality Control

Max Marks: 60

Date:

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		
A) Difference between line measurement and end measurement	CO1	06
B) Explain the need of measurement, possible error in measurement and precautions to be taken to eliminate errors.	CO1	06
C) Define/Explain following terms, i) Accuracy and Precision ii) Measurement and Calibration iii) Standard and Limit	CO1	06
Q.2 Solve Any Two of the following.		
A) Draw a neat sketch of sigma comparator and explain its working	CO2	06
B) What are various angle measurement methods. Explain any one with the help of neat sketch	CO2	06
C) Explain the working of pneumatic comparators with a neat sketch state its advantages and limitations	CO2	06
Q. 3 Solve Any Two of the following.		
A) State the principle of interferometer and explain how it is used in checking flatness and angle	CO2	06
B) Define surface roughness and explain CLA method for measurement of surface roughness	CO2	06
C) Sketch different types of texture and direction of lay obtained during measuring operation	CO3	06
Q.4 Solve Any Two of the following.		
A) What are different methods for measurement of pitch diameter. Explain two wire method with neat sketch	CO4	06
B) What are the different methods for the measurement for the measurement of effective diameter of screw thread	CO4	06
C) Different errors in screw threads	CO4	06
Q. 5 Solve Any Two of the following.		
A) Difference between quality control and quality assurance	CO5	06
B) Explain the concept of balance between cost of quality and value of quality	CO6	06

***** End *****